

A Technical Introduction to Naive Bayes classifier

$$P(C_k | x) \propto P(C_k) \prod_{i=1}^n P(x_i | C_k)$$

Section 1

To estimate the parameters for a feature's distribution, one must assume a distribution or generate nonparametric models for the features from the training set. The assumptions on distributions of features are called the "event model" of the naive Bayes classifier. For discrete features like the ones encountered in document classification (include spam filtering), multinomial and Bernoulli distributions are popular. These assumptions lead to two distinct models, which are often confused.

Section 2

$$\frac{p(S | D)}{p(\neg S | D)} = \frac{p(S)}{p(\neg S)} \prod_i \frac{p(w_i | S)}{p(w_i | \neg S)} \quad \left\{ \frac{p(S)}{p(\neg S)} \prod_i \frac{p(w_i | S)}{p(w_i | \neg S)} \right\}$$

Section 3

(This technique of "log-likelihood ratios" is a common technique in statistics. In the case of two mutually exclusive alternatives (such as this example), the conversion of a log-likelihood ratio to a probability takes the form of a sigmoid curve: see logit for details.) Finally, the document can be classified as follows. It is spam if $\frac{p(S | D)}{p(\neg S | D)} > 1$ (i. e., $\ln \frac{p(S | D)}{p(\neg S | D)} > 0$), otherwise it is not spam.

Section 4

However, given the sample, the evidence is a constant and thus scales both posteriors equally. It therefore does not affect classification and can be ignored. The probability distribution for the sex of the sample can now be determined:

Section 5

The following example assumes equiprobable classes so that $P(\text{male}) = P(\text{female}) = 0.5$. This prior probability distribution might be based on prior knowledge of frequencies in the larger population or in the training set.

Section 6

Thus, the probability ratio $p(S | D) / p(\neg S | D)$ can be expressed in terms of a series of likelihood ratios. The actual probability $p(S | D)$ can be easily computed from log $(p(S | D) / p(\neg S | D))$ based on the observation that $p(S | D)$

$+ p(\neg S | D) = 1$. Taking the logarithm of all these ratios, one obtains:

Section 7

The link between the two can be seen by observing that the decision function for naive Bayes (in the binary case) can be rewritten as "predict class C_1 if the odds of $p(C_1 | x)$ exceed those of $p(C_2 | x)$ ". Expressing this in log-space gives:

Section 8

Gaussian naive Bayes When dealing with continuous data, a typical assumption is that the continuous values associated with each class are distributed according to a normal (or Gaussian) distribution. For example, suppose the training data contains a continuous attribute, x . The data is first segmented by the class, and then the mean and variance of x is computed in each class. Let μ_k be the mean of the values in x associated with class C_k , and let σ_k^2 be the Bessel corrected variance of the values in x associated with class C_k . Suppose one has collected some observation value v . Then, the probability density of v given a class C_k , i.e., $p(x = v | C_k)$, can be computed by plugging v into the equation for a normal distribution parameterized by μ_k and σ_k^2 . Formally,

Section 9

Disadvantages Depending on the implementation, Bayesian spam filtering may be susceptible to Bayesian poisoning, a technique used by spammers in an attempt to degrade the effectiveness of spam filters that rely on Bayesian filtering. A spammer practicing Bayesian poisoning will send out emails with large amounts of legitimate text (gathered from legitimate news or literary sources). Spammer tactics include insertion of random innocuous words that are not normally associated with spam, thereby decreasing the email's spam score, making it more likely to slip past a Bayesian spam filter. However, with (for example) Paul Graham's scheme only the most significant probabilities are used, so that padding the text out with non-spam-related words does not affect the detection probability significantly. Words that normally appear in large quantities in spam may also be transformed by spammers. For example, «Viagra» would be replaced with «Viaagra» or «V!agra» in the spam message. The recipient of the message can still read the changed words, but each of these words is met more rarely by the Bayesian filter, which hinders its learning process. As a general rule,

this spamming technique does not work very well, because the derived words end up recognized by the filter just like the normal ones. Another technique used to try to defeat Bayesian spam filters is to replace text with pictures, either directly included or linked. The whole text of the message, or some part of it, is replaced with a picture where the same text is "drawn". The spam filter is usually unable to analyze this picture, which would contain the sensitive words like «Viagra». However, since many mail clients disable the display of linked pictures for security reasons, the spammer sending links to distant pictures might reach fewer targets. Also, a picture's size in bytes is bigger than the equivalent text's size, so the spammer needs more bandwidth to send messages directly including pictures. Some filters are more inclined to decide that a message is spam if it has mostly graphical contents. A solution used by Google in its Gmail email system is to perform an OCR (Optical Character Recognition) on every mid to large size image, analyzing the text inside.

Section 10

$$\text{evidence} = P(\text{male}) p(\text{height} \mid \text{male}) p(\text{weight} \mid \text{male}) p(\text{foot size} \mid \text{male}) + P(\text{female}) p(\text{height} \mid \text{female}) p(\text{weight} \mid \text{female}) p(\text{foot size} \mid \text{female})$$